

SME SAGE — Hackathon Presentation Guide

AI-Powered SME Decision Intelligence Platform | Full Technical & Pitch Dossier

1. Executive Pitch & Problem Statement

The Problem: Small & Medium Enterprises (SMEs) struggle with data scattered across Excel, CSVs, POS systems, and billing software. Traditional Business Intelligence tools (Dashboards) only answer 'What happened?' without giving clear operational recommendations.

The Solution: **SME SAGE** acts as an AI Copilot that guides the SME owner through a complete decision lifecycle:
RAW DATA → UNDERSTAND → DETECT → EXPLAIN → RECOMMEND → SIMULATE → DECIDE → TRACK

2. Complete Technology Stack

Layer	Technology / Tool	Purpose / Role in Project
Frontend Framework	React 18 + Vite 6 + TypeScript 5.7	Single Page Application (SPA) runtime, fast HMR bundling & strict type safety
Styling & Icons	Tailwind CSS 3.4 + Lucide React	Utility-first dark UI design, glassmorphic layout, responsive grid & iconography
Data Visualization	Recharts 2.15 + Custom SVG Gauge	Interactive line charts, bar charts, donut breakdowns & 0-100 radial health score gauge
Backend Runtime	Node.js v20.18 + Express 4.21 + tsx	REST API server on port 5000 with CORS, JSON middleware & hot-reload tsx execution
Database Layer	SQLite WASM (sql.js) + Persistent DB	In-memory + file-backed SQLite database (sme_sage.db) with zero native compilation
File Processing	xlsx (SheetJS) + Multer	Parsing uploaded Excel (.xlsx/.xls) / CSV spreadsheets & disk storage
Intelligence Engines	Custom Math & Rule Engines	Health Score (0-100), Price Elasticity Simulator, Grounded AI Q&A; & Data Cleaner

3. Frontend Architecture & Page Structure

State Management & Services:

- **BusinessContext:** Global React Context storing current active SME profile, demo dataset state, business health score, and refresh triggers.
- **ToastContext:** Provides floating notifications for user feedback and simulation updates.
- **Unified API Client (api.ts):** Axios-style fetch service connecting frontend components to backend REST endpoints.

Core User Interfaces & Modules:

Page Module	Key Functions & Presentation Highlights
DashboardPage	Executive KPIs (Revenue, Profit, Margin, Stock Value), 0-100 Health Score radial gauge, monthly growth charts.

AnalyticsPage	Detailed revenue breakdown, category sales rank, profit margins by SKU, overhead expense spike detection.
InventoryPage	Stock Coverage in months (Stock / Demand), overstock warnings (Mechanical Keyboard), stockout alerts (Power Bank), replenishment PO modal.
SimulatorPage	What-If decision simulation with price/discount sliders, demand elasticity modeling, projected revenue/profit, and mandatory disclaimers.
AssistantPage	'Ask Your Business' natural language query engine grounded in database metrics (Logistics spike, Keyboard capital lockup).
DecisionsPage	Audit log tracking committed business decisions, scenario notes, and implementation timeline.
DataUploadPage	Drag-and-drop CSV/Excel file parser, automated data sanitization preview, null filling, and Data Quality Score (0-100%).
ReportsPage	Printable executive stakeholder dossier with health scores, financial breakdown, and strategic recommendations.

4. Backend Core Engines & Mathematical Models

A. Business Health Score Algorithm (0 - 100)

Calculates a holistic 100-point index using 5 weighted business pillars:

Health Score = 0.25 × RevGrowth + 0.25 × Profitability + 0.20 × InventoryHealth + 0.15 × CustomerRetention + 0.15 × ExpenseControl

UrbanKart Retail Score: 78 / 100 (Stable business health, inventory efficiency requires optimization).

B. Stock Coverage Formula

Stock Coverage (Months) = Current Stock / Average Monthly Demand

- **Overstock Threshold (> 4.0 months):** Mechanical Keyboard = 5.1 months (■6.30 Lakhs capital locked up).
- **Low Stock Threshold (< 1.0 month):** Power Bank = 0.3 months (Stock = 18 units, 9 days runway remaining).

C. Price Elasticity & What-If Simulator Engine

Models demand shift based on price changes:

- **Demand Change (ΔQ%):** -Elasticity × (ΔPrice / Original Price)
- **Projected Revenue:** New Demand × New Price
- **Projected Profit:** Revenue - (Demand × Unit Cost) - Fixed Overhead

D. Grounded Q&A; Assistant Engine

Answers natural language questions by fetching pre-calculated SQL metrics to eliminate AI hallucination:

- 'Why did my profit decrease?' → Flags +21.1% logistics fee surge & -22% keyboard slump.
- 'Which product should I restock?' → Recommends 80-unit Power Bank reorder PO.

5. Demo Dataset: UrbanKart Retail (Consumer Electronics)

Product SKU	Status / Category	Key Business Finding
Wireless Earbuds	Growth Leader	Sales surged +31%, top revenue generator at ■2.06 Lakhs.
Mechanical Keyboard	Capital Lockup	Sales slumping (-22%), 420 units on hand (5.1 months coverage), ■6.30 Lakhs frozen.
Power Bank	Stockout Alert	18 units remaining (below reorder limit 30), stockout risk in 9 days.
Smart Watch	Steady Earner	Consistent performance with a strong 40% profit margin.
Logistics Courier Fees	Expense Spike	Logistics expense surged +21.1% in recent month due to express surcharges.

6. 10-Step Hackathon Demo Pitch Script

- **Step 1:** Open <http://localhost:3000>. Point out the sleek dark dashboard and Continuous Decision Pipeline concept.
- **Step 2:** Click 'LOAD DEMO SME' to instantly load UrbanKart Retail data.
- **Step 3:** Highlight the 78/100 Business Health Score and overall monthly profit breakdown.
- **Step 4:** Navigate to AI Insights. Show the 5 auto-generated diagnostic cards (Logistics surge +21%, Keyboard overstock).
- **Step 5:** Click 'SIMULATE THIS ACTION' on Mechanical Keyboard or Earbuds.
- **Step 6:** In Decision Simulator, adjust the discount slider to show live projected revenue, profit impact, and disclaimer.
- **Step 7:** Click 'APPLY SCENARIO' to save the simulation into the formal Decision Audit Log.
- **Step 8:** Navigate to Ask Your Business. Ask 'Why did my profit decrease this month?' to show grounded AI analysis.

- **Step 9:** Navigate to Inventory. Explain the Stock Coverage formula and generate a Power Bank Reorder PO.
- **Step 10:** Navigate to Reports. Show the printable executive dossier for bank/investor presentation.